# DS8001 Project-First Design: Focused, Defensible, and Finishable

**Approval verdict**

The topic is an excellent fit, but the present proposal should not be submitted unchanged.

**Recommended core**

Exact brute force + truncated-permutation scan + MI-File, with random versus farthest-first pivots and one exact BallTree baseline.

**Cut from committed scope**

MPT, prefix indexing, IVF, Euclidean second geometry, readmission prediction, and exhaustive testing at every N.

The guidelines explicitly favour **a careful study of a small number of well-motivated algorithms**, require substantive self-implemented algorithmic work, and identify exact-versus-approximate nearest-neighbour search as an appropriate topic. They also require experiments to answer the stated questions under fair, reproducible conditions. The grading weights reinforce this: Methods and Experiments are each 25%, Analysis is 20%, while the proposal is judged on clarity, feasibility, algorithmic focus, and appropriateness.

## 1. Immediate approval blockers

| Current choice | Decision | Why it risks approval or later marks |
|---|---|---|
| "The mixed distance is a proper metric" | **Rewrite conditionally and prove it** | Pairwise omission of missing attributes can destroy metric properties; Gower's original paper explicitly warns that missing values can affect desirable matrix properties. [jstor.org] |
| MPT as a "scoring refinement" | **Remove from committed scope** | MPT is a distinct **quantized-ranking representation and data structure**, not merely a scoring variation. [link.springer.com], [discover.gcu.ac.uk] The present proposal does not define its quantizer, stored cells, search procedure, or cost. |
| MIF + MPT + prefix index + trees + IVF + prediction | **Narrow aggressively** | This conflicts with the guideline's depth-over-breadth principle and makes feasibility difficult to defend. |
| KD-tree, Ball-tree and IVF as comparable baselines | **Keep only BallTree, under the identical metric** | Scikit-learn trees do not accept arbitrary Python callables, sparse input forces brute force, and IVF would require a separate Euclidean geometry. [scikit-learn.org], [scikit-learn.org] |

| Current choice | Decision | Why it risks approval or later marks |
|---|---|---|
| Intrinsic dimensionality claim | **Replace with measured distance concentration and pruning effectiveness** | Increasing $N$ does not manipulate intrinsic dimensionality. A causal claim would not follow from one dataset. |
| $N = 100{,}000$ after patient-held-out splitting | **Use 10k, 50k and the full available index partition** | The complete dataset has 101,766 encounters; reserving whole patients for validation and test necessarily leaves fewer than 100k index encounters. [archive.ics.uci.edu] |
| "Exact reranking" | **Clarify** | Reranking is exact only *within the selected candidate set*; the overall result remains approximate. |

**Bottom line:** the strongest submission is not "brute force + MIF + MPT." It is **brute force + a linear truncated-permutation method + an MI-File inverted implementation**. That gives two related but algorithmically distinct approximate methods and avoids promising an inadequately specified MPT.

---

# 2. Final research questions

## RQ1 — Parameter effects

> **At fixed data and metric, how do pivot count $n$, retained permutation length $L$, and reranking budget $c$—including their interactions—affect recall@K, query work, and index size?**

Directly answered by controlled sweeps at the largest index size.

## RQ2 — Access structure and pivot choice

> **At matched permutation representation and reranking budget, when does MI-File provide a better recall–work trade-off than scanning all truncated permutation signatures, and how does this depend on random versus farthest-first pivots?**

This cleanly isolates:

1. the quality of the permutation surrogate;
2. the additional approximation caused by inverted candidate generation; and
3. pivot selection.

Pivot-selection literature finds that random selection is often outperformed, but also that no strategy is universally best across permutation indexes. Farthest-first traversal is especially valuable academically because it is simple, self-implemented, costs $O(nM)$over a pivot-selection pool of size $M$, and has a 2-approximation interpretation for metric $k$-centre coverage—not a guarantee of ANN recall. [falchi.isti.cnr.it] [falchi.isti.cnr.it], [sites.cs.ucsb.edu]

### RQ3 — Scaling and geometric difficulty

> **As index size increases, how do build cost, query work, memory and recall scale, and how are pruning effectiveness and candidate requirements associated with measured distance concentration?**

Use observed diagnostics rather than an undefined latent "intrinsic dimension":

- coefficient of variation of query-to-index distances;
- relative contrast $\left(\bar{d} - d_K\right)/\bar{d}$;
- BallTree distance calls divided by $N$;
- MI-File candidate fraction required to reach selected recall targets.

Do not claim that concentration *causes* performance changes; report associations.

---

## 3. Algorithms to implement

Permutation indexing represents each object by pivots ordered by distance. Spearman-footrule-style rank differences then act as a surrogate for the original distance. [falchi.isti.cnr.it], [studylib.net]

### A. Exact brute-force kNN — self-implemented

For each query:

1. compute the mixed distance to every indexed encounter;
2. select the $K$smallest using partial selection;
3. order by `(distance, encounter_id)`.

This supplies ground truth and the exact distance-cost baseline.

- Query distances: $N$
- Query time: $O(Nf + N + K \log K)$, where $f$is feature-distance cost
- Additional index: none

## B. Truncated Permutation Scan (TPS) — self-implemented

For pivots $R = r_1, \ldots, r_n$, rank them by distance from object $x$. Define the truncated rank

$$\rho_L^x(r) = \begin{cases} \text{rank}_x(r), & \text{rank}_x(r) \le L, \\ L+1, & \text{otherwise.} \end{cases}$$

The surrogate score is

$$F_L(q,x) = \sum_r \in R |\rho_L^q(r) - \rho_L^x(r)|.$$

The absent-pivot rank $L+1$removes the ambiguity in the current proposal. Similar top-list rank treatment is used in permutation-index literature. [falchi.isti.cnr.it]

Search:

1. compute $n$query-to-pivot distances;
2. compute $F_L$for all $N$stored signatures;
3. take the best $c$surrogate candidates;
4. evaluate their true mixed distances;
5. return the exact top $K$inside that candidate set.

TPS is approximate because the surrogate may fail to place a true neighbour in the first $c$.

## C. Truncated MI-File — self-implemented

For each pivot $r$, store an inverted posting list of `(object_id, retained_rank)` for objects whose rank is at most $L$. This matches the core MI-File organization described in the literature. [falchi.isti.cnr.it], [studylib.net]

Search:

1. compute the query prefix;
2. read the posting lists of its retained pivots;
3. form the union of objects sharing at least one retained pivot;
4. compute the **same** $F_L$score for those candidates;
5. take the best $c$, evaluate true distances, and return top $K$.

This comparison is unusually clean: **TPS and MI-File use the same pivots, signatures, surrogate, candidate budget, queries and true metric.** MI-File's benefit must therefore come from avoiding signature scans; its extra risk is that zero-overlap objects cannot enter its candidate set.

Measure zero-overlap failures explicitly: the proportion of true top-$K$ neighbours absent before reranking.

### D. Exact BallTree — library baseline

The proposed mixed metric can be transformed exactly into weighted Manhattan coordinates:

- numeric/ordinal scalar coordinates are scaled by their feature weight and fixed scale;
- each nominal feature becomes one-hot coordinates scaled by half its weight, because L1 distance between two distinct one-hot values is 2;
- missing indicators receive their own weighted coordinate.

A BallTree using `metric="manhattan"` therefore searches the **same geometry**, not a second Euclidean representation. Scikit-learn lists Manhattan as supported and exposes the number of distance calls for tree queries. [scikit-learn.org], [github.com]

Validate exactness against brute force on synthetic cases and sampled real queries before using it. Keep **BallTree only**; adding KDTree would not materially strengthen the questions.

---

## 4. Defensible mixed-type metric

Use six conceptual groups with equal total group weight:

1. age;
2. encounter/utilization counts;
3. admission, discharge and specialty;
4. diagnoses;
5. laboratory results;
6. medications and treatment.

Weights within a group sum to one. Thus the 23 medication fields collectively receive the same influence as another domain group rather than 23 times its influence.

For numeric feature $j$, let $m_j(x)$indicate missingness and let $z_j(x)$be the observed value or a fixed training-partition median when missing:

$$d_j(x,y) = \lambda_j \mid m_j(x) - m_j(y) \mid + \frac{\mid z_j(x) - z_j(y) \mid}{s_j},$$

where $s_j > 0$is frozen from the index/training partition. Drop zero-scale features. No clipping is required.

- **Ordinal:** use absolute differences between declared ordered ranks, with the same missing-indicator construction.
- **Nominal:** discrete distance $1\[x_j \neq y_j]$; `UNKNOWN`, `OTHER`, and clinically meaningful `NOT_MEASURED` are fixed states.
- **Diagnoses:** deterministically map `diag_1–diag_3` into documented ICD-9 chapters rather than treating every raw code mismatch identically. ICD-9 supplies explicit chapter ranges such as circulatory 390–459 and respiratory 460–519. [icd9data.com]
- **A1C/max glucose:** `NOT_MEASURED` is a meaningful state, distinct from an unknown value. The published dataset documentation explicitly distinguishes result ranges from tests not taken.
- **Exclude:** `encounter_id`, `patient_nbr`, `readmitted`, derived readmission labels, race, gender and weight. IDs and labels would distort or leak information; protected demographics are unnecessary for the algorithmic research question.

With fixed positive scales, fixed nonnegative pair-independent weights, a fixed denominator and no pairwise feature deletion, each component is an L1 or discrete metric; their weighted sum is therefore a metric. Conventional Gower pairwise omission should not be used: later work demonstrates that missing-value omission can produce unattainable similarities. [annals-csis.org]

### Metric sensitivity

Commit to only one sensitivity:

- **Primary:** equal total weight per domain group.
- **Sensitivity:** equal weight per retained column.

Do **not** add Euclidean/one-hot as a second research geometry. The exact L1 encoding already supports BallTree without changing neighbours.

---

# 5. Complete experimental protocol

| Element | Final specification |
|---|---|
| Split | Group patients into 90% index/training, 5% validation-query, 5% test-query using one fixed seed. No patient crosses partitions. |
| Queries | Sample one encounter per patient: 300 validation queries and 500 final test queries. |
| Preprocessing | Fit scales, medians, category maps, diagnosis maps and weights on the index/training partition only. |
| Scaling sets | Nested prefixes of a deterministic shuffle: 10k, 50k and the complete index partition; report its actual size. |

| Element | Final specification |
|---|---|
| Ground truth | Cache brute-force top-50 results for all validation/test queries at each nested $N$. |
| $K$ | $K = 10$primary; $K = 50$sensitivity only. |
| Pivots | $n \in 32,64,128$. Select from a fixed 10k index-only pivot pool. |
| Prefix | $L \in 8,16,32$, retaining only valid $L \leq n$settings. |
| Reranking | c/N\in{0.1%,0.5%,1%,5%}, with $c \geq K$. |
| Pivot methods | Uniform random and farthest-first; five seeds each. Smaller $n$uses prefixes of the corresponding 128-pivot sequence. |
| Tuning | Sweep the grid only on validation at full $N$; choose three Pareto operating points per method—low cost, balanced, high recall. |
| Final testing | Freeze those configurations, then evaluate once on test at all three $N$values. |
| Threads | Single-threaded unless every method receives the same allowance; record environment variables and versions. |
| Ties | Deterministic `(distance, encounter_id)` ordering. Report strict recall and tie-aware recall where any item with distance $\leq d_K$counts as relevant. |
| Duplicates | Count exact preprocessed duplicates; retain them, but report a sensitivity allowing at most one neighbour per patient. |

The official rubric specifically expects splitting/tuning protocols, repeated randomized experiments, fair conditions, running time, memory, scalability and reproducibility.

---

# 6. Evaluation and analysis

## Primary metrics—promise these in the proposal

- recall@10 against exact brute force;
- query-to-pivot distances;
- true reranking distances;
- TPS signature entries scored;
- MI-File postings read, distinct accumulators touched and candidate count;
- serialized index bytes;
- build distances and build time;
- median query latency;
- scaling with $N$.

No single work counter is sufficient: a true mixed distance is more expensive than an integer posting update. Present separate counts plus latency.

### Final-report additions

- 95% patient-level bootstrap confidence intervals;
- paired per-query differences using identical queries;
- separate variation across pivot seeds and across queries;
- p95 latency and peak build/query memory;
- strict versus tie-aware recall;
- group-weight sensitivity;
- distance-concentration and pruning diagnostics;
- Pareto fronts for recall versus distance evaluations, total work, latency and bytes.

The analysis should explain *where approximation is lost*:

1. surrogate-ranking loss in TPS;
2. zero-overlap/candidate-generation loss added by MI-File;
3. reranking-budget loss;
4. changes caused by pivot coverage;
5. tree-pruning deterioration when distances concentrate.

---

# 7. Quantitative feasibility

Using 100k as a conservative upper bound:

| Work | Approximate count |
|---|---|
| Brute-force validation + test ground truth | $800 \times 100k = 80$million distances |
| One 128-pivot index build | 12.8 million distances |
| Five random + five farthest-first maximum pivot sets | 128 million pivot distances |
| Query-to-pivot work across those sets | about 1.0 million distances |
| Largest reranking timing run, $c = 5$ | 2.5 million true distances per selected test configuration |
| Cached 800×100k float32 distance matrix | about 320 MB |
| One 100k×128 float32 pivot table | about 51 MB |
| Packed MI-File at $L = 32$ | 3.2 million postings; roughly 16–19 MB before metadata |

This is realistic **if** you:

- vectorize/chunk the mixed distance kernel;

- process one pivot seed at a time;
- cache the maximum 128-pivot table and derive smaller $n, L$;
- reuse cached validation ground-truth distances for reranking during tuning;
- time selected configurations with fresh distance computation;
- avoid Python loops over record pairs.

---

# 8. Threats to validity worth retaining

- **Repeated encounters:** patient-grouped queries prevent leakage; one-neighbour-per-patient sensitivity tests domination by frequent patients.
- **ICD-9 simplification:** chapter grouping loses clinical detail; acknowledge this rather than claiming clinical similarity.
- **Medication dominance:** controlled by group weights.
- **Missingness:** `NOT_MEASURED` and unknown are not conflated.
- **Ties and duplicates:** deterministic and tie-aware reporting.
- **Limited scale:** say "scaling within the available dataset," not proof of web-scale behaviour.
- **No clinical claim:** the data are a challenging mixed-type retrieval benchmark; the study evaluates algorithms, not clinical decisions.

---

# 9. Two-student division

- **Jared Smouter:** preprocessing/metric implementation, brute-force oracle, exact L1 transformation, BallTree validation, metric and leakage tests.
- **Amarnadh Nalla:** permutation signatures, TPS, MI-File, farthest-first pivots, index serialization and operation counters.
- **Joint:** experiment configuration, reproducibility scripts, validation selection, statistical analysis, plots, report and cross-review.

---

# Proposal-writing checklist

## Must appear on the one page

1. **Empirical Analysis path and focused title**
   **Recommended title:**

*Permutation-Based Approximate kNN Search for Mixed-Type Hospital Encounters: Signature Scan versus MI-File*

2. **One-sentence problem**

   We study how truncated permutation indexes trade recall, query work and memory against exact kNN on a mixed-type hospital-encounter benchmark; no clinical-decision claim is made.

3. **The three research questions above**, shortened but preserving matched conditions.
4. **Self-implemented components**

   We implement brute-force exact kNN, truncated permutation signatures with top-list Spearman-footrule scoring, an MI-File inverted candidate generator, exact reranking within candidate sets, and random and farthest-first pivot selection.

5. **One comparable library baseline**

   A scikit-learn BallTree using an exactly equivalent weighted-Manhattan encoding provides an exact tree baseline under the same distance.

6. **Defensible metric statement**

   Our fixed weighted mixed metric uses scaled absolute differences for numeric/ordinal attributes, discrete mismatch for nominal attributes, explicit missingness states, and equal total weights across clinical feature groups; preprocessing is fitted only on index patients.

7. **Data and leakage control**

   Patients are grouped into index/training, validation-query and final test-query partitions; IDs and readmission outcomes are excluded from similarity features.

8. **Concrete experiment promise**

   On nested 10k, 50k and full-index instances, we tune $n$, $L$and candidate budget on validation patients, freeze Pareto configurations, and evaluate 500 test queries over five pivot seeds.

9. **Core metrics**

Recall@10, true distances, pivot distances, postings/signatures processed, candidates, index bytes, build cost, latency, memory and scaling, with paired uncertainty summaries.

10. **Complexity promise**

    The report analyses $O(Nn)$pivot distances plus per-object ranking, $O(NL)$stored signatures/postings, query pivot cost, surrogate work and $c$true reranking distances.

# Remove from the current proposal

- "MPT scoring refinement"
- prefix index
- IVF
- KD-tree plus Ball-tree
- Euclidean second representation
- readmission-prediction extension
- undefined intrinsic dimensionality claim
- "proper metric" without conditions
- "large-scale scalability"
- "exact reranking" without "within the candidate set"

# Details that can wait for the final report

- full feature-by-feature preprocessing table;
- exact serialized data types;
- complete validation grid;
- bootstrap implementation;
- p95 latency protocol;
- duplicate counts and observed tie rates;
- all diagnostic plots;
- hardware and package-version table.

**Final recommendation:** submit a proposal promising **TPS versus MI-File under one proven mixed metric**, with brute force and exact BallTree as controls and farthest-first pivot selection as the theoretical secondary factor. That is clearer, more feasible, more algorithmically coherent, and better aligned with the 25% Methods, 25% Experiments and 20% Analysis criteria than the current broad benchmark.